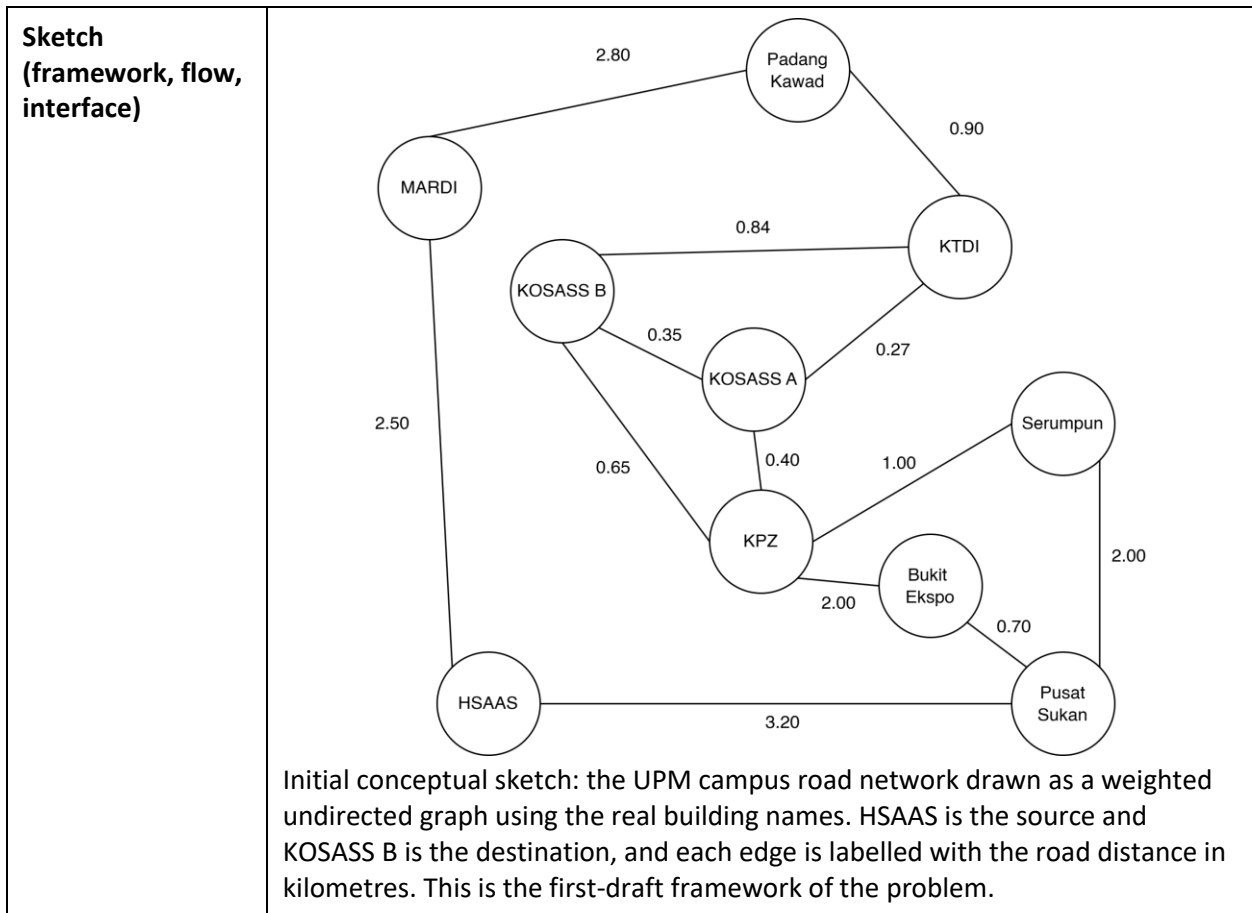


Initial Project Plan (week 10, submission date: 5 June 2026)

Group Name	Group 10		
Members	Name	Matric Number	Email
	Muhammad Irfan Daniel Bin Md Sham	226089	226089@student.upm.edu.my
	Muhammad Aiman Haziq Bin Mohd Salleh	222707	222707@student.upm.edu.my
	Muhammad Aidil Qayyim Bin Din	223555	223555@student.upm.edu.my
	Zahin Zulqarnain Bin Zulkifli	222292	222292@student.upm.edu.my
Problem scenario description	A fire breaks out at KOSASS B in Universiti Putra Malaysia (UPM) due to an electrical fault at 1:00 PM on a weekday (peak hour). At the same time, a football tournament at the field in front of KTDI causes heavy traffic congestion on the nearby roads. An ambulance must be dispatched from Hospital Sultan Abdul Aziz Shah (HSAAS) to KOSASS B as quickly as possible, but several candidate routes are affected by the congestion, making it difficult to choose the fastest route manually.		
Why it is important	In an emergency, every extra minute of ambulance travel can worsen the condition of injured students (e.g. smoke inhalation). Manually guessing the shortest route is unreliable because road conditions keep changing due to the fire and the tournament. An optimal routing algorithm guarantees the ambulance takes the lowest-cost route, accounting for both road distance and traffic congestion.		
Problem specification	The problem is modelled as a Single-Source Shortest Path problem on a weighted undirected graph $G = (V, E, W)$. Locations are vertices (10 nodes) and roads are edges (13 edges), where each edge weight is the effective travel cost = distance (km) $\times$ traffic multiplier. All edge weights are non-negative. The goal is to minimise the total path cost from the source HSAAS to the destination KOSASS B: minimise $\sum w(v_i, v_{i+1})$.		
Potential solutions	Four shortest-path / graph approaches were reviewed: Breadth-First Search (BFS), Floyd-Warshall, Greedy Best-First Search, and Dijkstra's Algorithm. Dijkstra's Algorithm was selected because it fully accounts for edge weights, targets the single-source problem, and guarantees the optimal shortest path for graphs with non-negative weights.		



Project Proposal Refinement (week 11, submission date: 12 June 2026)

Group Name	Group 10	
Members	Name	Role
	Muhammad Irfan Daniel Bin Md Sham	Problem Statement & Scenario (Section 2.0); Specification of an Algorithm (Section 4.0)
	Muhammad Aiman Haziq Bin Mohd Salleh	Implementation (Section 5.0)
	Muhammad Aidil Qayyim Bin Din	Introduction (Section 1.0); Algorithm Verification (Section 6.0)
	Zahin Zulqarnain Bin Zulkifli	Model Development for the Scenario (Section 3.0); Analysis of the Algorithm (Section 7.0)

	All members	Conclusion (Section 8.0), online portfolio and final presentation
Problem statement	Determine the optimal (lowest-cost) route for an ambulance travelling from Hospital Sultan Abdul Aziz Shah (HSAAS) to KOSASS B in UPM during a fire emergency, taking into account both road distance and traffic congestion, so that the ambulance reaches the scene in the shortest possible travel time.	
Objectives	1. To model the UPM campus road network as a weighted undirected graph where each edge weight represents the effective travel cost (distance × traffic multiplier). 2. To apply Dijkstra's Algorithm to determine the shortest path from HSAAS to KOSASS B. 3. To verify the correctness of the algorithm through manual calculation, recurrence analysis, and asymptotic verification. 4. To analyse the time and space complexity of the algorithm using asymptotic notation.	
Expected output	1. The optimal route from HSAAS to KOSASS B: A → C → F → H → J (HSAAS → Pusat Sukan → Bukit Ekspo → KPZ → KOSASS B). 2. The total effective travel cost of the optimal route (6.55). 3. Execution time of the algorithm as evidence of its efficiency.	
Problem scenario description	A fire breaks out at KOSASS B in Universiti Putra Malaysia (UPM) due to an electrical fault at 1:00 PM on a weekday (peak hour). At the same time, a football tournament at the field in front of KTDI causes heavy traffic congestion on the nearby roads. An ambulance must be dispatched from Hospital Sultan Abdul Aziz Shah (HSAAS) to KOSASS B as quickly as possible, but several candidate routes are affected by the congestion, making it difficult to choose the fastest route manually.	
Why it is important	In an emergency, every extra minute of ambulance travel can worsen the condition of injured students (e.g. smoke inhalation). Manually guessing the shortest route is unreliable because road conditions keep changing due to the fire and the tournament. An optimal routing algorithm guarantees the ambulance takes the lowest-cost route, accounting for both road distance and traffic congestion.	
Problem specification	The problem is modelled as a Single-Source Shortest Path problem on a weighted undirected graph $G = (V, E, W)$. Locations are vertices (10 nodes) and roads are edges (13 edges), where each edge weight is the effective travel cost = distance (km) × traffic multiplier. All edge weights are non-negative. The goal is to minimise the total path cost from the source HSAAS to the destination KOSASS B: minimise $\sum w(v_i, v_{i+1})$.	
Potential solutions	Four shortest-path / graph approaches were reviewed: Breadth-First Search (BFS), Floyd-Warshall, Greedy Best-First Search, and Dijkstra's Algorithm. Dijkstra's Algorithm was selected because it fully accounts for edge weights,	

	targets the single-source problem, and guarantees the optimal shortest path for graphs with non-negative weights.							
Sketch (framework, flow, interface)	Refined model: the campus graph is abstracted into a formal weighted graph where each location is relabelled as a single letter (A = HSAAS source, J = KOSASS B destination) and each edge now carries the effective travel cost. This finalised representation is exactly what the Dijkstra program reads as input. Example output from code is the same result as the graph: <pre>Shortest effective cost from A (HSAAS) to every node A -> A : 0.00 A -> B : 2.50 A -> C : 3.20 A -> D : 5.30 A -> E : 5.20 A -> F : 3.90 A -> G : 6.20 A -> H : 5.90 A -> I : 6.30 A -> J : 6.55 Source node : A (HSAAS) Destination node : J (KOSASS B) Optimal route : A -> C -> F -> H -> J Total cost : 6.55 Time taken : 2386300 nanoseconds : 2.3863 milliseconds</pre>							
Methodology	<table><tr><th>Milestone</th><th>Time</th></tr><tr><td>Scenario formulation and problem definition: define the UPM fire emergency scenario, source/destination, and the weighted-graph model. Initial UI/flow sketching.</td><td>wk10</td></tr><tr><td>Algorithm paradigm comparison (BFS, Floyd-Warshall, Greedy Best-First, Dijkstra); selection and justification of Dijkstra's Algorithm; defining methodology.</td><td>wk11</td></tr></table>	Milestone	Time	Scenario formulation and problem definition: define the UPM fire emergency scenario, source/destination, and the weighted-graph model. Initial UI/flow sketching.	wk10	Algorithm paradigm comparison (BFS, Floyd-Warshall, Greedy Best-First, Dijkstra); selection and justification of Dijkstra's Algorithm; defining methodology.	wk11	
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	Java implementation: build the adjacency list and min-priority queue, code the core Dijkstra logic, then test and debug.	wk12
	Conduct correctness verification (manual trace, recurrence, asymptotic) and analyse time complexity $O((V+E) \log V)$ and space complexity $O(V+E)$.	wk13
	Prepare the online portfolio, compile documentation, and structure the final presentation deck.	wk14

Project Progress (Week 10 – Week 14)

Milestone 1	Planning, Modelling & Algorithm Selection	
Date (week)	Week 10 – Week 11	
Description / sketch	Finalised the UPM ambulance routing scenario (fire at KOSASS B, congestion caused by the football tournament near KTDI). Modelled the campus road network as a weighted undirected graph of 10 nodes and 13 edges using effective travel cost (distance × traffic multiplier). Reviewed candidate algorithms (BFS, Floyd-Warshall, Greedy Best-First Search and Dijkstra) and selected Dijkstra's Algorithm. Submitted the initial project plan and refined proposal.	
Role	Muhammad Aidil Qayyim Bin Din	Wrote the introduction framing the emergency-routing problem (Section 1.0); planned the correctness-verification approach (Section 6.0).
	Muhammad Irfan Daniel Bin Md Sham	Defined the problem statement and disaster scenario, including geographical setting and impact (Section 2.0); reviewed candidate algorithms and justified Dijkstra (Section 4.0).
	Zahin Zulqarnain Bin Zulkifli	Developed the graph model, covering data type, objective function, constraints and example dataset (Section 3.0); outlined the complexity-analysis framework (Section 7.0).

	<table> <tr> <td>Muhammad Aiman Haziq Bin Mohd Salleh</td><td>Drafted the program flow and initial pseudocode for the implementation (Section 5.0).</td></tr> </table>	Muhammad Aiman Haziq Bin Mohd Salleh	Drafted the program flow and initial pseudocode for the implementation (Section 5.0).
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Milestone 2	Implementation, Verification, Analysis & Documentation	
Date (week)	Week 12 – Week 14	
Description / sketch	Implemented Dijkstra's Algorithm in Java using an adjacency list and a min-priority queue. Tested the program on the campus graph and obtained the optimal route $A \rightarrow C \rightarrow F \rightarrow H \rightarrow J$ with a total effective travel cost of 6.55. Verified correctness through manual calculation, recurrence analysis and asymptotic verification, analysed the time complexity $O((V+E) \log V)$ and space complexity $O(V+E)$, built the online portfolio, and prepared the final presentation.	
Role	Muhammad Aiman Haziq Bin Mohd Salleh	Wrote, debugged and tested the Java program; produced the flowchart, pseudocode and program output (Section 5.0).
	Muhammad Aidil Qayyim Bin Din	Completed the correctness verification, recurrence analysis and asymptotic verification (Section 6.0).
	Zahin Zulqarnain Bin Zulkifli	Completed the time- and space-complexity analysis using asymptotic notation (Section 7.0).
	Muhammad Irfan Daniel Bin Md Sham	Finalised the algorithm specification and comparison table; refined the scenario write-up (Sections 2.0 & 4.0).
	All members	Jointly wrote the conclusion (Section 8.0), built the online portfolio and prepared the final presentation.